

Training Only the First Layer : Feature Learning in a Three-Layer MLP

Hugo Vidal

The goal of this note is to understand how updating the first layer W_1 of a MLP affects the features representation of the second layer $\mathbf{h}_2$, and more specifically, how this impacts the correlation between the prediction $\hat{f}$ and the target function f^* . To this end, this note proceeds in two steps:

1. Derive the evolution of the second-layer features $\mathbf{h}_2$ induced by a learning step on the first layer W_1 .
2. Characterize how such a change in $\mathbf{h}_2$ affects the correlation between $\hat{f}$ and f^* .

Model definition :

We consider a three-layer MLP defined by :

$$f(x) = \mathbf{w}_3^\top \sigma(W_2 \sigma(W_1 x))$$

with the following dimensions :

$$x \in \mathbb{R}^d, \quad W_1 \in \mathbb{R}^{d_1 \times d}, \quad W_2 \in \mathbb{R}^{d_2 \times d_1}, \quad \mathbf{w}_3 \in \mathbb{R}^{d_2}.$$

Second-layer preactivation :

We define the **preactivation** of the second layer for neuron i as :

$$h_{2,i}(x) = \mathbf{w}_{2,i}^\top \sigma(W_1 x)$$

where $\mathbf{w}_{2,i}$ is the i -th row of matrix W_2 .

Feature evolution after a training step :

If we train the j -th layer of the model slowly enough, we can perform a first-order Taylor expansion of the i -th neuron of the second layer $h_{2,i}$:

$$\Delta_{W_j} h_{2,i}(x) = \langle \nabla_{W_j} h_{2,i}(x), \Delta W_j \rangle$$

We now attempt to compute the explicit form of $\Delta_{W_j} h_{2,i}(x)$ induced by training of the second, then the first layer of the network.

I. Evolution of second-layer features during an update of W_2

1. Gradient of preactivation $h_{2,i}$ with respect to W_2

$$\begin{aligned} \nabla_{W_2} h_{2,i}(x) &= \begin{pmatrix} \vdots \\ \cdots \frac{\partial}{\partial W_2^{k,l}} \sum_m^{d_1} W_2^{i,m} \sigma(W_1 x)_m \cdots \\ \vdots \end{pmatrix}, \quad k \in [d_2], l \in [d_1] \\ &= \begin{pmatrix} 0 \\ \vdots \\ \cdots \sigma(W_1 x)_l \cdots \\ \vdots \\ 0 \end{pmatrix} \end{aligned}$$

Therefore:

$$\nabla_{W_2} h_{2,i}(x) = \begin{pmatrix} 0 & \cdots & 0 & \sigma(W_1 x) & 0 & \cdots & 0 \end{pmatrix}$$

$$\nabla_{W_2} h_{2,i}(x) \in \mathbb{R}^{d_2 \times d_1}$$

$\nabla_{W_2} h_{2,i}(x)$ is not zero at the i -th row.

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2. Expression of ΔW_2

$$\Delta W_2 = -\frac{\eta}{n} \nabla_{W_2} \sum_{q=1}^n \ell(\mathbf{x}_q)$$

Assuming that the output layer weights are initialized small, the correlation loss is a good approximation of the square loss :

$$\begin{aligned} &\approx \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) \cdot \nabla_{W_2} \hat{f}(\mathbf{x}_q) \\ &= \frac{\eta}{n} \nabla_{W_2} \hat{f}(X) f^*(X) \end{aligned}$$

where $X \in \mathbb{R}^{d \times n}$ is the matrix of the n data points.

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3. Gradient of the output with respect to W_2

$$\begin{aligned}\nabla_{W_2} \hat{f}(X) &= \nabla_{W_2} (\mathbf{w}_3^\top \sigma(W_2 \sigma(W_1 X))) = \sum_{j=1}^{d_2} w_{3,j} \nabla_{W_2} \sigma(\mathbf{w}_{2,j}^\top \sigma(W_1 X)) \\ &= \begin{pmatrix} \vdots \\ \cdots \sum_j^{d_2} w_{3,j} \sigma'(h_{2,j}(X)) \frac{\partial}{\partial W_2^{k,l}} \sum_m^{d_1} W_2^{j,m} \sigma(W_1 X)_m \cdots \\ \vdots \end{pmatrix} \\ &= \begin{pmatrix} \vdots \\ \cdots w_{3,k} \sigma'(h_{2,k}(X)) \sigma(W_1 X)_l \cdots \\ \vdots \end{pmatrix}\end{aligned}$$

Hence in compact form :

$$\nabla_{W_2} \hat{f}(X) = (\mathbf{w}_3 \odot \sigma'(\mathbf{h}_2(X))) \odot \sigma(W_1 X) \in \mathbb{R}^{d_2 \times d_1 \times n}$$

4. Increment of preactivations $h_{2,i}$ during training of W_2

$$\Delta_{W_2} h_{2,i}(x) = \langle (0, \dots, \sigma(W_1 x), \dots, 0), \Delta W_2 \rangle = \sum_{\ell=1}^{d_1} \sigma(W_1 x)_\ell \Delta_{W_2} W_{2,i,\ell}$$

Substituting $\Delta W_{2,i,\ell}$:

$$\begin{aligned}&= \sum_{q=1}^n \sum_{\ell=1}^{d_1} \sigma(W_1 x)_\ell \frac{\eta}{n} w_{3,i} \sigma'(h_{2,i}(\mathbf{x}_q)) \sigma(W_1 \mathbf{x}_q)_\ell f^*(\mathbf{x}_q) \\ \Delta_{W_2} h_{2,i}(x) &= \frac{\eta}{n} \sum_{q=1}^n [w_{3,i} \sigma'(h_{2,i}(\mathbf{x}_q)) \sigma(W_1 x)^\top \sigma(W_1 \mathbf{x}_q)] f^*(\mathbf{x}_q)\end{aligned}$$

Hence :

$$\Delta_{W_2} \mathbf{h}_2(x) = \frac{\eta}{n} \sum_{q=1}^n [(\mathbf{w}_3 \odot \sigma'(\mathbf{h}_2(\mathbf{x}_q))) \sigma(W_1 x)^\top \sigma(W_1 \mathbf{x}_q)] f^*(\mathbf{x}_q)$$

$$\Delta_{W_2} \mathbf{h}_2(x) = \frac{\eta}{n} \sum_{q=1}^n [\nabla_{\mathbf{h}_2(\mathbf{x}_q)} \hat{f}(\mathbf{x}_q) \cdot \sigma(W_1 x)^\top \sigma(W_1 \mathbf{x}_q)] f^*(\mathbf{x}_q)$$

II. Evolution of second-layer features during an update of W_1

1. Gradient of preactivation $h_{2,i}$ with respect to W_1

We compute :

$$\begin{aligned}
\nabla_{W_1} h_{2,i}(x) &= \begin{pmatrix} \vdots \\ \cdots \frac{\partial}{\partial W_1^{k,l}} \sum_m^{d_1} W_2^{i,m} \sigma(W_1 x)_m \cdots \\ \vdots \end{pmatrix}, \quad k \in [d_1], l \in [d] \\
&= \begin{pmatrix} \vdots \\ \cdots \sum_m^{d_1} W_2^{i,m} \sigma'(W_1 x)_m \sum_p^d \frac{\partial}{\partial W_1^{k,l}} W_1^{m,p} x_p \cdots \\ \vdots \end{pmatrix} \\
&= \begin{pmatrix} \vdots \\ \cdots W_2^{i,k} \sigma'(W_1 x)_k x_l \cdots \\ \vdots \end{pmatrix}
\end{aligned}$$

In compact form :

$$\nabla_{W_1} h_{2,i}(x) = \mathbf{w}_2^i \odot \sigma'(W_1 x) \odot x \in \mathbb{R}^{d_1 \times d}$$

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2. Expression of ΔW_1

We consider the weight update :

$$\Delta W_1 = -\frac{\eta}{n} \nabla_{W_1} \sum_{q=1}^n \ell(\mathbf{x}_q)$$

Assuming the output weights are initialized small, we approximate the loss as :

$$\Delta W_1 = -\eta \nabla_{\theta} \mathcal{L}(\hat{f}_{\theta}, \mathcal{D}) \approx \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) \cdot \nabla_{W_1} \hat{f}(\mathbf{x}_q) = \frac{\eta}{n} \nabla_{W_1} \hat{f}(X) f^*(X)$$

where $X \in \mathbb{R}^{d \times n}$ is the matrix of training samples.

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3. Gradient of the output with respect to W_1

We compute :

$$\nabla_{W_1} \hat{f}(X) = \nabla_{W_1} (\mathbf{w}_3^{\top} \sigma(W_2 \sigma(W_1 X))) = \sum_{j=1}^{d_2} w_{3,j} \nabla_{W_1} \sigma(\mathbf{w}_{2,j}^{\top} \sigma(W_1 X))$$

Unfolding dimensions :

$$\nabla_{W_1} \hat{f}(X) = \begin{pmatrix} \vdots \\ \cdots \sum_j^{d_2} w_{3,j} \frac{\partial \sigma(h_{2,j}(X))}{\partial W_1^{k,l}} \cdots \\ \vdots \end{pmatrix}, \quad k \in [d_1], l \in [d]$$

$$\begin{aligned}
&= \begin{pmatrix} \vdots \\ \cdots \sum_j^{d_2} w_{3,j} \sigma'(h_{2,j}(X)) \frac{\partial h_{2,j}(X)}{\partial W_1^{k,l}} \cdots \\ \vdots \end{pmatrix} \\
&= (\mathbf{w}_3 \odot \sigma'(\mathbf{h}_2(X)))^\top (\nabla_{W_1} \mathbf{h}_2(X))
\end{aligned}$$

So :

$$\nabla_{W_1} \hat{f}(X) = (\mathbf{w}_3 \odot \sigma'(\mathbf{h}_2(X)))^\top (W_2 \odot \sigma'(W_1 X) \odot X) \in \mathbb{R}^{d_1 \times d \times n}$$

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4. Increment of preactivations $h_{2,i}$ during training of W_1

$$\begin{aligned}
\Delta_{W_1} h_{2,i}(x) &= \langle \nabla_{W_1} h_{2,i}(x), \Delta W_1 \rangle = \left\langle \nabla_{W_1} h_{2,i}(x), \frac{\eta}{n} \nabla_{W_1} \hat{f}(X) \cdot f^*(X) \right\rangle \\
&= \frac{\eta}{n} \sum_{q=1}^n \langle \nabla_{W_1} h_{2,i}(x), \nabla_{W_1} \hat{f}(\mathbf{x}_q) \rangle f^*(\mathbf{x}_q) \\
&= \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) \sum_{j,k,\ell}^{d_2, d_1, d} W_2^{i,k} \sigma'(W_1 x)_{k\ell} w_{3,j} \sigma'(h_{2,j}(\mathbf{x}_q)) W_2^{j,k} \sigma'(W_1 \mathbf{x}_q)_k x_{\ell} \mathbf{x}_{\ell,q} \\
&= \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) \sum_{j=1}^{d_2} w_{3,j} \sigma'(h_{2,j}(\mathbf{x}_q)) \sum_{k=1}^{d_1} W_2^{j,k} \sigma'(W_1 x)_k W_2^{i,k} \sigma'(W_1 \mathbf{x}_q)_k \sum_{\ell=1}^d x_{\ell} \mathbf{x}_{\ell,q} \\
&= \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) (x^\top \mathbf{x}_q) \sum_{j=1}^{d_2} w_{3,j} \sigma'(h_{2,j}(\mathbf{x}_q)) \sum_{k=1}^{d_1} W_2^{j,k} \sigma'(W_1 x)_k W_2^{i,k} \sigma'(W_1 \mathbf{x}_q)_k \\
&= \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) (x^\top \mathbf{x}_q) \sum_{j=1}^{d_2} w_{3,j} \sigma'(h_{2,j}(\mathbf{x}_q)) (W_2^j \odot \sigma'(W_1 \mathbf{x}_q))^\top (W_2^i \odot \sigma'(W_1 x)) \\
&= \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) (x^\top \mathbf{x}_q) [\mathbf{w}_3 \odot \sigma'(\mathbf{h}_2(\mathbf{x}_q))]^\top [(W_2 \odot \sigma'(W_1 \mathbf{x}_q)) \cdot (W_2^i \odot \sigma'(W_1 x))]
\end{aligned}$$

$$\Delta_{W_1} h_{2,i}(x) = \frac{\eta}{n} \sum_{q=1}^n f^*(\mathbf{x}_q) (x^\top \mathbf{x}_q) [\nabla_{\mathbf{h}_2(\mathbf{x}_q)} \hat{f}(\mathbf{x}_q)]^\top [\nabla_{\mathbf{h}_1(\mathbf{x}_q)} \mathbf{h}_2(\mathbf{x}_q) \cdot \nabla_{\mathbf{h}_1(x)} h_{2,i}(x)]$$

$$\Delta_{W_1} h_{2,i}(x) = -\frac{\eta}{n} \sum_{q=1}^n (x^\top \mathbf{x}_q) \nabla_{\hat{f}(\mathbf{x}_q)} \ell(\mathbf{x}_q) [\nabla_{\mathbf{h}_2(\mathbf{x}_q)} \hat{f}(\mathbf{x}_q)]^\top [\nabla_{\mathbf{h}_1(\mathbf{x}_q)} \mathbf{h}_2(\mathbf{x}_q) \cdot \nabla_{\mathbf{h}_1(x)} h_{2,i}(x)]$$

$$\Delta_{W_1} \mathbf{h}_2(x) = -\frac{\eta}{n} \sum_{q=1}^n (x^\top \mathbf{x}_q) \nabla_{\hat{f}(\mathbf{x}_q)} \ell(\mathbf{x}_q) [\nabla_{\mathbf{h}_2(\mathbf{x}_q)} \hat{f}(\mathbf{x}_q)]^\top [\nabla_{\mathbf{h}_1(\mathbf{x}_q)} \mathbf{h}_2(\mathbf{x}_q) \cdot \nabla_{\mathbf{h}_1(x)} \mathbf{h}_2(x)^\top]$$

Interpretation: The expression for $\Delta_{W_1} h_{2,i}(x)$ indicates that the update of $\mathbf{h}_2(x)$ depends on a combination of three factors:

1. the similarity between x and training samples $\mathbf{x}_q$ via the term $x^\top \mathbf{x}_q$.
2. the alignment between the direction of feature change $\nabla_{\mathbf{h}_1(x)} h_{2,i}(x)$ and the direction in which the loss decreases on each $\mathbf{x}_q$.
3. the way the first-layer weights W_1 evolve to reduce the loss via the $\nabla_{\mathbf{h}_1(x)}$.

In particular, the term $\nabla_{\mathbf{h}_1(\mathbf{x}_q)} \mathbf{h}_2(\mathbf{x}_q)$ captures how sensitive $\mathbf{h}_2$ is to changes in the representation $\mathbf{h}_1$, which depends on the evolution of W_1 . Therefore, $\mathbf{h}_2(x)$ evolves more in directions where modifying $\mathbf{h}_1$ can simultaneously help reduce the loss on the first and second layers. This implies that the update of $\mathbf{h}_2$ is not random: over training, the evolution of $\mathbf{h}_2(x)$ is fostered toward a value that minimizes the loss over neighborhoods of similar examples, modulated by the other constraints on the learning of W_1 .

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Questions:

Does this interpretation appropriately capture the role of W_1 in shaping the evolution of $\mathbf{h}_2(x)$? Can this expression be understood as evidence that $\Delta_{W_1} \mathbf{h}_2(x)$ evolves in a non-random direction? Are other insights to extract from this expression?

Next steps :

1. See the evolution of the correlation between $\hat{f}$ and f^* when making a $\Delta_{W_1} \mathbf{h}_2$ step.
2. If this doesn't work, try to find a way to show that a $\Delta_{W_1} \mathbf{h}_2$ step contributes to the learning of the task by the MLP.